

Udit Jain

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SUMMARY

Results-driven Software Development Engineer and M.Tech (CSE) graduate from NIT Warangal (Graduating 2026) with experience at Intel, Cisco and C-DOT. Proficient in Java, Spring Boot, and ReactJS; passionate about building scalable, reliable, and maintainable systems through innovation and collaboration.

PROFESSIONAL EXPERIENCE

Intel Corporation - Bengaluru, Software Engineer Intern Jul 2025 – May 2026

- Re-engineered the platform's query pipelines to support complex, multi-conditional data discovery; built an automated, real-time sorting mechanism that **accelerated high-volume data access by ~40%** for cross-functional teams' reaching the latest platform rollouts and data streams.
- Developed an automated metrics aggregation engine from scratch, consolidating complex product statistics into a single, high-precision reporting system; integrated strict backend calculation logic that **eliminated ~8 hours of manual reporting weekly** across multiple global Business Units.
- Led a platform-wide UI/UX optimization to abstract complex legacy alphanumeric identifiers into intuitive, user-friendly aliases; removed operational friction, **cut data navigation errors by ~30%, and reclaimed ~20 engineering hours/month** for internal teams.

CISCO Systems - Bengaluru, Software Engineer Intern May 2025 - Jul 2025

- Constructed an AI-driven automation system using LangChain and LLMs to automate software patch deployment, simulation, and report generation for router hardware systems, **reducing manual intervention by 70%**.
- Devised the overall system design to seamlessly integrate Jenkins execution, patch loading, and log analysis into a single unified flow, **collapsing 6+ manual steps into one pipeline and cutting deployment turnaround time by ~50%**.
- Programmed a scalable Webex chatbot with **four intelligent agents**, accelerating regression testing and **reducing debugging time by 38%**.
- Applied multithreading to create parallel agent instances, enabling concurrent automation tasks and **refining overall system throughput by 25%**.

Centre for Development of Telematics (C-DOT) - Delhi, Software Engineer Jan 2023 - Jul 2024

- Led full-stack feature development by applying design patterns and scalable architecture principles of the NCCS platform homepage using Spring Boot, ReactJS, MySQL, and Thymeleaf, **raising load speed by 35%** and enhancing accessibility for internal government users.
- Implemented modular RESTful APIs using Object-Oriented Programming (OOP) principles and caching strategies, **trimming average API response time by ~30%** and boosting backend efficiency.
- Collaborated with a **12-member cross-functional team** to ship complex production-ready features with near-zero post-deployment issues.
- Contributed to hardware automation in the POTP project by validating ROADM5 configurations and **debugging 30+ processor cards** (82xx, 85xx, T1022), **elevating system uptime and trimming setup time by 25%**.

Infosys Technologies Private Limited - Bengaluru, Digital Specialist Engineer Jul 2022 - Dec 2022

- Designed and deployed a scalable e-commerce web platform using the MERN stack, **serving 10,000+ daily users** for a major client.
- Enhanced system performance by ~42%** by redesigning the ReactJS frontend and optimizing Node.js and MongoDB pipelines.
- Coordinated with backend and DevOps teams to ensure reliable API integrations, version control via Git, and CI/CD automation, **compressing release cycles by ~30%** and lifting deployment frequency.

SKILLS

- | | |
|--------------------------|---|
| Languages & Databases | - Core Java, Java 8, JavaScript, Python, SQL (MySQL), NoSQL (MongoDB) |
| Frameworks & Libraries | - Spring Boot, ReactJS, LangChain |
| OS, Tools & Technologies | - Git, Jenkins, Postman, REST APIs, Linux, Windows, macOS |
| CS Foundations | - System Design, OOPs, Data Structures, Algorithms |

PROJECTS

TrustX-Defect, Explainable Software Defect Prediction Pipeline

Tech Stack - Python, XGBoost/Random Forest, SHAP, Scikit-Learn, Pandas, Matplotlib

- Engineered an **advanced Explainable AI (XAI) pipeline** across 12 cross-project transfer configurations using PROMISE benchmarking datasets to evaluate model generalization under severe distribution shifts.
- Introduced 3 novel behavior-grounded metrics—**Gradient-of-Loss Reliability (GLR)**, **Expected Calibration Error (ECE)**, and **-Perturbation Hit@k**—synthesizing them into a unified **Reliability Score (RS)** framework to formally evaluate XAI trustworthiness.
- Formulated a reliability-aware optimization strategy using train-only preprocessing, probability calibration, and threshold tuning; significantly stabilized prediction precision and achieved near-perfect feature causal relevance (**0.9998 Hit@k**) in cross-project prediction.

NeuroPredict, Parkinson's Disease Detector

Tech Stack - Python, CatBoost, XGBoost, Scikit-Learn, NumPy, Pandas, Matplotlib

- Independently architected a machine learning model to detect Parkinson's disease from vocal biomarkers using **22+ extracted acoustic features** (jitter, shimmer, NHR).
- Achieved **96.6% accuracy** and **91.4% MCC**, outperforming baseline models; optimized via **SMOTE balancing** and **GridSearchCV** for **precision >94%**.
- Built a fully automated ML pipeline with cross-validation and result visualization for reproducible outcomes, **reducing experiment setup time by ~50%** and **ensuring 100% reproducible runs**; packaged the project via a virtual environment for local deployment and QA.

CERTIFICATIONS

Amazon ML Summer School 2025

- Completed intensive training on **machine learning, deep learning, and production-grade ML systems**.
- Gained hands-on exposure to **supervised and unsupervised learning, model evaluation, model optimization, and large-scale ML applications**.
- Strengthened foundations in **AI model design, data preprocessing, and performance tuning** for real-world deployment scenarios at Amazon.

EDUCATION

M.Tech. - Computer Science and Engineering Aug 2024 - May 2026

National Institute of Technology, Warangal

CGPA - 8.33

B.Tech. - Computer Science and Engineering Aug 2018 - Jul 2022

Maharaja Agrasen Institute of Technology, New Delhi

CGPA - 9.06